

$$\tan \theta = F_V / F_H$$

$$\square = 39^\circ, \text{ anticlockwise above horizontal.}$$

C1

A1

- 4 (a) (i)** The internal energy of a substance is the sum of the random distribution of kinetic and potential energies associated with the molecules of a system. **B1**

- (ii)** At the same temperature, water vapour and water have the same total kinetic energy of molecules but water vapour has higher total potential energy of molecules (due to weaker intermolecular forces of attraction) **B1**

Hence the internal energy per unit mass of water vapour is higher than that of water **B1**

- (b) (i)**

$$\frac{pV}{T} = nR = \text{constant}$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

$$\frac{101000(15000)}{288} = \frac{890 \times V_2}{228}$$

C1

$$V_2 = 1.3476 \times 10^6$$

$$= 1.35 \times 10^6 \text{ m}^3$$

A1

$$\begin{aligned}
 \text{(ii)} \quad KE_{avg} &= \frac{3}{2} kT \\
 &= \frac{3}{2} (1.38 \times 10^{-23}) (228) \\
 &= 4.72 \times 10^{-21} \text{ J}
 \end{aligned}$$

B1

$$\begin{aligned}
 \text{(iii)} \quad \Delta U &= U_f - U_i \\
 &= \frac{3}{2} p_f V_f - \frac{3}{2} p_i V_i \\
 &= \frac{3}{2} [890(1.3476 \times 10^6) - (101000)(15000)] \\
 &= -4.7345 \times 10^8 \\
 &= -4.73 \times 10^8 \text{ J}
 \end{aligned}$$

C1

A1